

# Learned variable-support priors accelerate symbolic regression of physical laws

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## Abstract

Symbolic regression—the recovery of closed-form equations from data—is a powerful route to interpretable physical laws, but its cost grows steeply with the number of candidate variables and operators. In a typical measurement table only a few columns and a few operators actually appear in the governing equation, yet a solver must still pay for every irrelevant branch of the search tree. Here we show that a learned, observation-conditioned *front-end* that predicts which variables are relevant—a *support prior*—substantially improves and accelerates symbolic regression without changing the backend solver. The front-end couples a graph-attention network over a variable co-occurrence graph with a statistical-feature model, and recovers the true variables with recall 1.000 and near-perfect per-task recall across three physics suites, so it essentially never discards a real variable. Supplied to the evolutionary backend PySR under a fixed time budget, the prior doubles exact-law recovery (from 30% to 60%), reduces the search space by 86%, and accelerates time-to-solution by 3–6 $\times$  on typical formulas; on black-box tabular benchmarks it raises held-out predictive accuracy while a same-size random-support control degrades it, isolating learned guidance from dimensionality reduction. The approach is solver-agnostic, adds only  $\sim 40$  ms per task, and keeps final law selection with a verifiable symbolic backend.

## 1 Introduction

The automated discovery of governing equations from observation is a long-standing goal of the physical sciences<sup>1–3</sup>. Symbolic regression (SR) addresses it directly: rather than fitting an opaque function, SR searches the space of mathematical expressions for a compact formula that both fits the data and remains human-interpretable<sup>4,5</sup>. The resulting equation can be dimensionally checked, extrapolated and falsified, properties that black-box predictors lack. This interpretability has made SR central to data-driven physics, from rediscovering conservation laws to proposing new constitutive relations<sup>2,6,7</sup>.

The central obstacle is combinatorial. An SR backend searches over expression trees whose leaves are drawn from the input variables and whose internal nodes are drawn from an operator library; the number of candidate trees grows exponentially with the number of admissible leaves and operators and with tree depth<sup>8,9</sup>. In physics problems this cost is largely wasted: a governing law typically involves only a handful of the measured variables and a small subset of the operator library, yet a solver run over all columns and operators still explores an enormous number of branches that cannot appear in the answer. Existing systems attack this with dimensional analysis, problem decomposition, sparsity priors or efficient enumeration<sup>2,6,10,11</sup>, but the raw width of the search space remains the dominant cost under any fixed compute budget.

We take a complementary route: instead of designing a faster search, we *narrow the space the search must explore*. We frame search-space reduction as a standalone, reusable front-end problem (Fig. 1). Given a small observation set  $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^q$  with  $q$  as small as 50–100, a learned model predicts which input columns (and, as an auxiliary, which operators) are likely to participate in the governing law, and passes this *support prior* to a conventional SR backend. The backend then performs the final symbolic search and model selection over the reduced space, so the output remains a verifiable equation rather than a neural approximation.

This division of labour is deliberate and, we argue, the correct one given the current state of neural SR. Models that decode the full equation directly from data have advanced rapidly<sup>12–16</sup>, but they remain unreliable at *assignment*: they often recover the right variables and operators yet place them in the wrong terms, so exact symbolic equivalence stays low. Predicting *support*, by contrast, is a far easier and more robust target, and even a perfect-recall variable filter hands a search backend an exponential head start while never making the true law unreachable. Our contributions are: (i) we cast variable- and operator-support prediction as a reusable, backend-agnostic SR front-end; (ii) we present a graph-attention support predictor, ensembled with a statistical-feature model, that attains recall 1.000 across three physics suites; and (iii) we quantify the downstream effect on a fixed-budget backend—doubled exact recovery, 86% search-space reduction and 3–6 $\times$  speedups—against a same-size random-support control that isolates learned guidance from dimensionality reduction.

## 2 Related work

**Symbolic regression backends.** Genetic-programming search over expression trees remains the workhorse of SR<sup>1,4</sup>, with PySR providing a fast, configurable modern implementation whose variable and operator sets can be restricted externally<sup>5</sup>. Physics-tailored systems add inductive structure: AI-Feynman exploits dimensional analysis and recursive decomposition<sup>2</sup>, SINDy imposes sparsity over a library of candidate terms<sup>6,10</sup>, dimensional-constraint search restricts units<sup>7</sup>, and parallel symbolic enumeration shares sub-expressions for efficiency<sup>11</sup>. Our front-end is complementary to all of these: it emits a reduced variable/operator set that any such backend can consume.

**Neural equation models.** Transformer and reinforcement-learning approaches learn distributions over expressions and decode them from numerical input<sup>9,12–14,17</sup>. Hybrid planners couple a neural proposal with tree search<sup>15,16</sup>, multimodal pre-training aligns equation tokens with sampled numeric behaviour<sup>18,19</sup>, and e-graph and diffusion methods offer alternative ways to represent or generate equivalent symbolic forms<sup>20,21</sup>; generative latent-space search has also been proposed<sup>22</sup>. These methods motivate our front-end framing: the shared, reliable signal they extract is *which ingredients matter*, even when full end-to-end decoding remains brittle.

**Physics-aware neural modelling.** A parallel line builds physical structure into neural models as an inductive bias—Hamiltonian and Lagrangian networks<sup>23,24</sup>, neural differential equations<sup>25</sup>, symbolic distillation of trained networks<sup>26</sup>, and concept-driven or abductive law-discovery systems<sup>27–29</sup>. These support our emphasis on keeping the final expression symbolic and verifiable while using the neural component only to narrow the search.

**Representations, benchmarks and generalisation.** Because the front-end consumes *sets* of observations, permutation-invariant and set-attention architectures are natural building blocks<sup>30,31</sup>; the contrastive and latent components of representation learning that prevent collapse or encode hierarchy are also relevant<sup>32–34</sup>. We evaluate with the SRBench protocol<sup>8</sup> and PMLB datasets<sup>35</sup>, and connect to recent studies of pre-training generalisation, LLM-assisted discovery and scientific latent spaces<sup>36–39</sup>. Our reported comparison is deliberately controlled: a learned support prior improves a fixed backend under matched full-search and same-size random-support baselines.

### 3 Results

#### The front-end recovers every true physical variable

We first evaluate the support prior intrinsically, without any backend. For each task the front-end receives  $q = 100$  noisy observations whose columns include the genuine variables together with random “distractor” columns, and must predict the true variable subset. Across three suites of physical laws—AI-Feynman (118 equations), a curated real-law benchmark (real591, 91 tasks) and an extended physics-law set (1085 equations)—the predictor attains recall 1.000 (or 0.999) with perfect recall on essentially every task (Fig. 2). In other words it almost never drops a variable that the law uses; its only error mode is occasionally retaining a few extra noise columns, which merely costs the backend a little additional search rather than making the law unreachable. Precision ranges from 0.85 to 1.00, reflecting an intentional high-recall operating point: removing a true variable is fatal to recovery, whereas keeping a spurious one is recoverable downstream. For reference, a naive statistical baseline trained on few records with name-matched labels reaches only recall 0.48, confirming that the graph-attention model contributes the decisive signal.

#### A perfect-recall prior strengthens a fixed-budget solver

We next insert the front-end before PySR<sup>5</sup> and compare three conditions at a fixed 10 s per-task budget on a 30-equation Feynman subset with  $\sim 20$  distractor columns: (A) *full PySR* over all columns and the full 17-operator library; (B) *+variable prior*, restricting the columns to the predicted support; and (C) *+operator prior*, additionally restricting the operator library. Restricting variables raises exact-law recovery (held-out  $R^2 \geq 0.999$ ) from 9/30 (30%) to 18/30 (60%)—a +30 percentage-point gain—while lifting mean held-out  $R^2$  from 0.919 to 0.967, reducing the average column count from 23.2 to 3.2 (an 86% reduction) and halving the effective search vocabulary from 40 to 20 (Fig. 3). The variable prior thus converts a large fraction of budget-limited failures into exact recoveries simply by removing branches that could never appear in the law.

#### The gain is search guidance, not dimensionality reduction

A natural concern is that any reduction in column count would help. We control for this with a *same-size random-support* baseline that retains the same number of columns but selects them without using the observations. On 24 PMLB/SRBench black-box tabular tasks (three seeds,  $q = 100$ , 5 s budget), the learned support raises mean held-out  $R^2$  from 0.347 to 0.495 with 10 distractor variables and from 0.315 to 0.488 with 20 distractors, using only 31% and 18% of the input columns respectively; the same-size random support instead *degrades* performance to 0.220 and  $-0.028$  (Fig. 4). At matched column count the random control is far worse than full search, whereas the learned prior is better—direct evidence that the front-end supplies observation-conditioned information about *which* columns matter, not merely *how many*.

#### Pruning the search space accelerates time-to-solution

Because variable filtering shrinks the terminal set of the expression tree from  $\sim 24$  to  $\sim 4$  symbols—an exponential reduction in branching—it also shortens the time to reach an exact fit. Measuring time-to-solution with early stopping on exact recovery, the front-end yields  $3\text{--}6\times$  speedups on typical physics formulas (Fig. 5); in some cases it converts a within-budget failure of full search into a fast success. The exception is deep-tree formulas whose bottleneck is expression *depth* rather than variable count, where pruning the variable set gives little benefit. Front-end inference adds only

$\sim 40$  ms per task, near-constant in the number of observations and negligible against the seconds of backend search it saves.

### Operator priors are net-neutral but a useful safeguard

Restricting the operator library in addition to variables (condition C) does not improve mean exact recovery (it is slightly lower,  $14/30$ ), because operators are intrinsically harder to infer from small, noisy domains and a mis-removed operator is fatal. We therefore report operator prediction honestly as an ablation rather than a primary result. It nonetheless has a clear conditional value: when full-operator search returns a *pathological* high- $R^2$  fit built from exotic operators, restricting the library to the predicted operators reliably recovers a sane solution (Fig. 6; e.g. held-out  $R^2$  improving from 0.19 to 0.92). The triggering signatures—an exotic low-probability operator, anomalous complexity, or  $R^2$  stuck just below 1—are visible at inference without the ground truth, so an adaptive “re-run with operator restriction when the fit looks pathological” policy is deployable in practice.

## 4 Discussion

Our results support a simple thesis: for symbolic regression of physical laws, a learned *support prior* is a high-leverage, low-risk intervention. Because a perfect-recall variable filter never makes the true law unreachable, it can be applied unconditionally; because it removes an exponential number of irrelevant branches, it materially improves both the success rate and the speed of a fixed-budget backend. The front-end is solver-agnostic—its output is a reduced variable/operator set that any tree-search, enumeration or generative backend<sup>5,11,22</sup> can consume—and cheap enough ( $\sim 40$  ms) to prepend to every run. The same-size random-support control shows the benefit is genuine search guidance rather than mere dimensionality reduction, and the black-box results indicate that the prior helps even when no symbolic ground truth exists, by steering a fixed-budget solver toward the informative columns.

The honest limitation, shared with direct neural SR<sup>13,14</sup>, is that the neural component cannot yet *assign* variables and operators to the correct positions in the expression reliably; this is precisely why we leave final term assignment to a symbolic, verifiable backend rather than emitting the equation end-to-end. Closing this assignment gap—through richer compositional latents, explicit constant/exponent recovery, and contrastive objectives that keep the observation and formula representations aligned<sup>18,32</sup>—is, in our view, the central open problem in moving from a front-end prior toward reliable end-to-end neural discovery. Even short of that goal, the present front-end is immediately useful: it makes existing physical-law solvers both more successful and several times faster, with a provably safe failure mode, and it is compatible with the dimensional and physics-aware priors that already strengthen scientific SR<sup>2,7,26</sup>.

## 5 Methods

**Problem setup.** Each task provides  $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^q$  with  $\mathbf{x}_i \in \mathbb{R}^m$ , where only an unknown subset  $S^* \subseteq \{1, \dots, m\}$  of columns appears in the target law  $f^*$ . The front-end predicts  $\hat{S} \approx S^*$  (and an optional operator set  $\hat{O}$ ) and passes it to an SR backend run under a fixed time budget. We deliberately pad tasks with random distractor columns to emulate the realistic regime in which a measurement table contains many irrelevant channels.

**Front-end architecture.** The variable-support predictor is an ensemble of two complementary models. (1) A graph-attention network<sup>30,31</sup> operates on a graph whose nodes are the input columns;

node features combine per-column statistics (moments, correlation with the target, monotonicity and dimensional descriptors) and edges encode a variable co-occurrence prior learned from a knowledge graph of physical laws together with observation-derived correlation. Attention over this graph lets the model reason about which columns *jointly* support a plausible law rather than scoring columns in isolation. (2) A random forest scores each column from the same per-column statistical features. The final score is  $s(\text{col}) = \text{RF}(\text{col}) + \sigma(\text{GAT}(\text{col}))$ , and a column is retained if  $s \geq \tau$  with a low threshold ( $\tau = 0.10$ ) chosen to favour recall. An analogous graph-attention model with graph-level pooling produces the multi-label operator prediction used for the optional operator prior.

**Training data.** The predictors are trained only on internal physical-law records—formula templates with sampled observation tables drawn from a curated knowledge graph of physical laws—never on the external evaluation suites. Training pools of 8k–40k formulas were used, with observation tables pre-sampled across variable ranges and a 10–30 distractor curriculum so the front-end is robust to the number of irrelevant columns at test time. An adversarial (GAN–graph-attention) variant was explored but gave no consistent gain and is not used.

**Backend integration and evaluation.** We use PySR<sup>5</sup> as the downstream backend because it exposes variable and operator restrictions directly and is reproducible locally; the same prior interface applies to enumeration<sup>11</sup> and generative<sup>22</sup> backends. All comparisons fix the backend budget across conditions, so the only changed quantity is the search space exposed to the solver. *Exact recovery* denotes held-out  $R^2 \geq 0.999$ . Support quality is measured by precision and recall against the known law variables; the same-size random-support control retains the same number of columns selected without the front-end, isolating learned guidance from dimensionality reduction. Time-to-solution is measured with early stopping on exact recovery, and front-end overhead is measured separately. All reported numbers are produced by the released evaluation scripts from the recorded result artifacts.

**Data availability.** The external benchmarks (AI-Feynman, PMLB/SRBench) are public<sup>2,8,35</sup>; the physical-law training records are materialised from public formula and range information. The recorded result artifacts that back every table and figure are released with the code.

**Code availability.** Code for the front-end, training, evaluation and figure generation, together with the result artifacts and the trained graph-attention weights, is released at <https://github.com/<anonymized>>.

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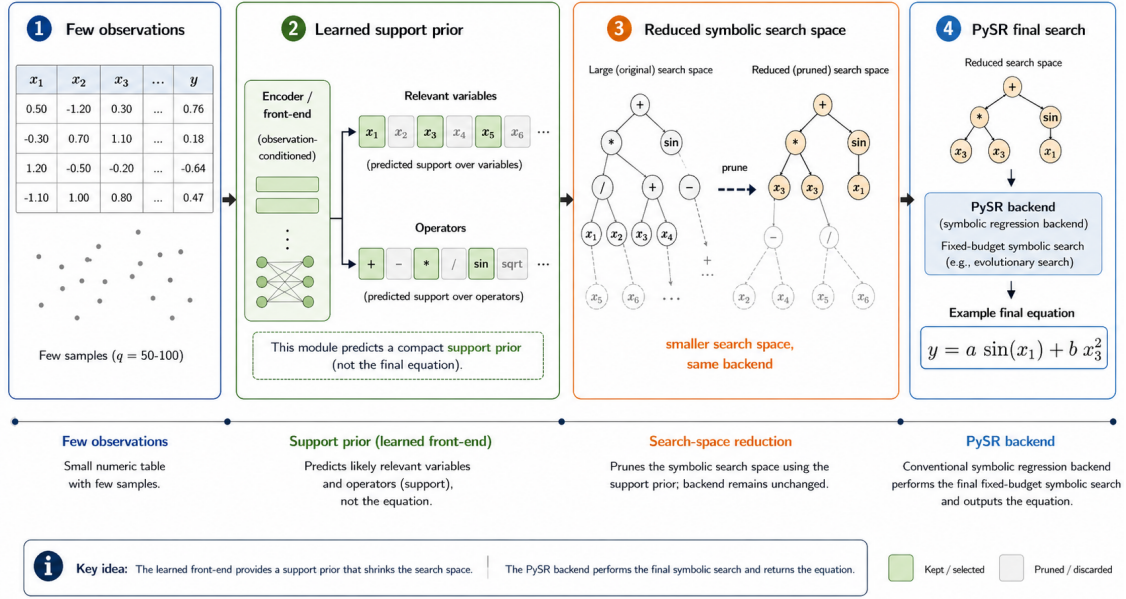
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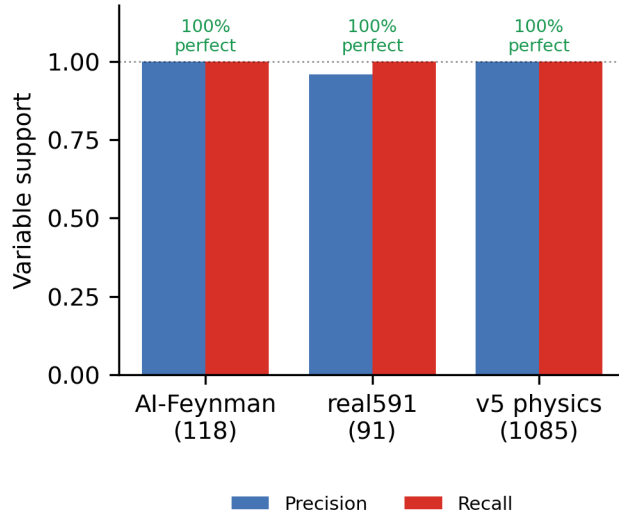
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# Figures

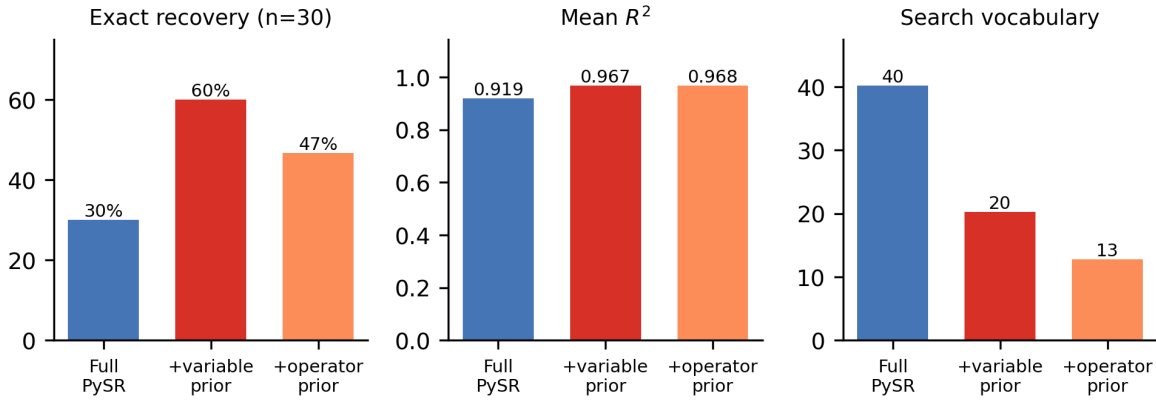


**Fig. 1. A learned front-end narrows the symbolic-regression search space.** From a small set of observations the front-end predicts a compact support prior (relevant variables and, optionally, operators); a conventional symbolic-regression backend then performs the final search over the reduced space and returns a verifiable equation. The neural model proposes *where* to look; the backend decides *what* the law is.

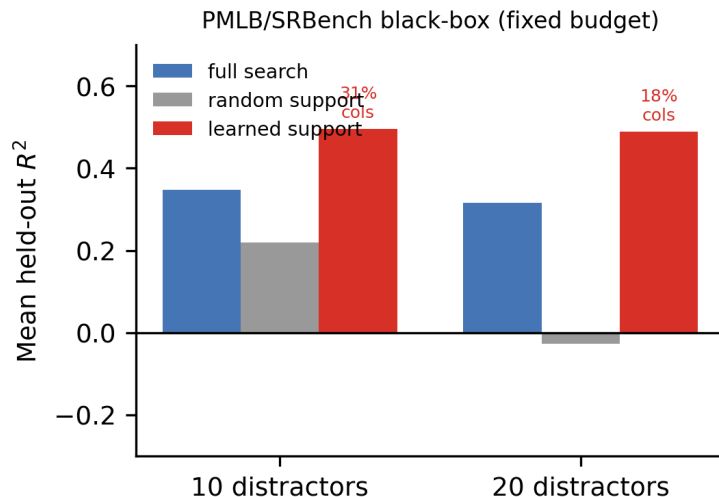




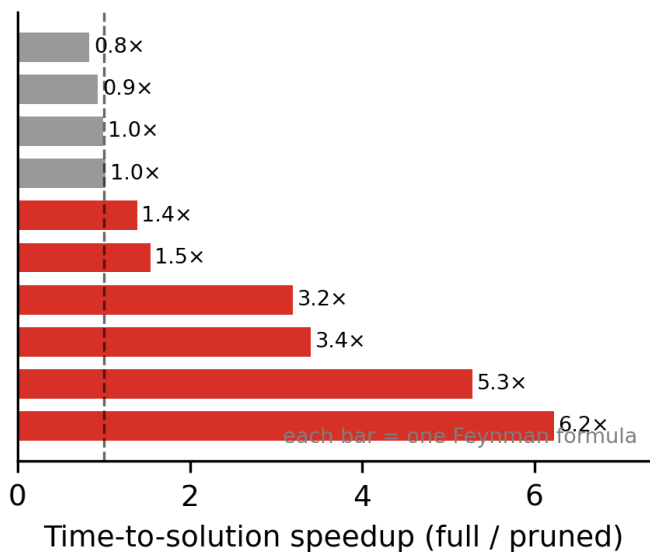
**Fig. 2. The front-end recovers every true variable.** Precision and recall of predicted variable support across three physics suites. Recall is 1.000 (AI-Feynman, real591) or 0.999 (extended set), with perfect recall on  $\sim 100\%$  of tasks, so a true variable is essentially never dropped; the high-recall bias trades a little precision for a safe failure mode.



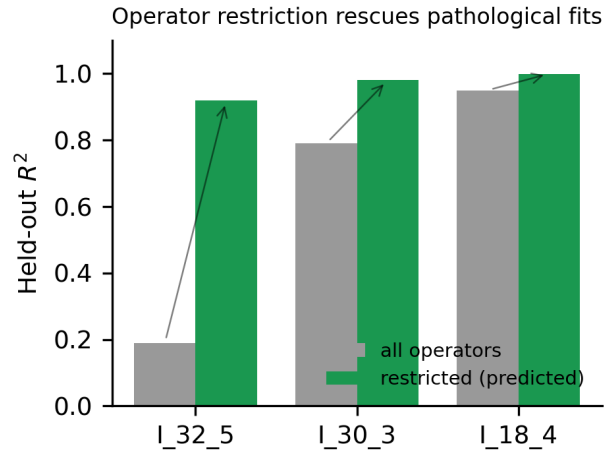
**Fig. 3. A variable prior strengthens a fixed-budget solver.** Full PySR versus +variable prior versus +operator prior on a 30-equation Feynman subset (10 s budget,  $\sim 20$  distractors). The variable prior doubles exact recovery (30%  $\rightarrow$  60%), raises mean  $R^2$  (0.919  $\rightarrow$  0.967) and halves the search vocabulary (40  $\rightarrow$  20); adding the operator prior cuts the vocabulary further but does not improve exact recovery.



**Fig. 4. The gain is search guidance, not fewer columns.** Mean held-out  $R^2$  on 24 PMLB/SRBench black-box tasks (three seeds, fixed 5 s budget) at 10 and 20 distractor variables. At matched, reduced column count (labels show columns relative to full search) the learned support *improves* over full search while a same-size random support *degrades* far below it.



**Fig. 5. Pruning the search space accelerates time-to-solution.** Per-formula speedup (full-search time divided by pruned-search time) with early stopping on exact recovery. Typical physics formulas solve 3–6 $\times$  faster; deep-tree formulas (depth-, not variable-limited) show little gain.



**Fig. 6. Operator restriction rescues pathological fits.** For cases where full-operator search returns a spurious high- $R^2$  solution built from exotic operators, restricting the library to the predicted operators recovers a sane fit (held-out  $R^2$  before versus after restriction). Operator priors are net-neutral on average but valuable as an adaptive safeguard.